

# Earth-State Embeddings: self-supervised representations of the observed ocean, probed against the Atlantic overturning circulation

blauewelt/earth — technical report, v1

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## Abstract

We train a small self-supervised model to compress everything the observing system knows about each point of the ocean, each month, into a single vector — an *embedding* — and then ask whether those vectors, which never see any transport measurement, carry the Atlantic Meridional Overturning Circulation (AMOC). They do: a linear read-out from the embeddings along  $26.5^\circ\text{N}$  attains a year-blocked cross-validated correlation of  $r = 0.60$  with the monthly, deseasonalized RAPID transport series — to our knowledge the first cross-validated monthly-scale figure against real RAPID data from inputs that never include RAPID. A temporal model over the frozen embeddings predicts the ocean’s anomaly field beyond a year ahead (anomaly correlation 0.62 at one month, crossing the conventional 0.5 useful-skill line near four months), while the transport read-out loses skill after about one season, consistent with its wind-driven monthly variance. We document data, methods, metrics (in the verification conventions of the forecasting literature), all experiments to date, and the honest baselines — including the sobering one: wind stress alone recovers  $r = 0.53$  of the monthly transport signal, so the embedding’s monthly edge over pure Ekman physics is  $+0.07$ , with the density-structure contribution appearing mainly at low frequencies and in event anatomy.

## 1 Introduction

The AMOC moves on the order of  $17\text{ Sv}$  of water northward through the Atlantic’s upper kilometre ( $1\text{ Sv} = 10^6\text{ m}^3\text{ s}^{-1}$ ), carries roughly a petawatt of heat, and is directly measured at only a handful of latitudes, for only about two decades. It is the canonical example of a climate-critical quantity that is *composite*: part wind-driven (Ekman transport), part density-driven (geostrophic shear), part boundary currents. No single satellite or float sees it; it must be inferred from the state of the ocean around it.

This project asks a representation-learning question of that inference problem: *if we compress the whole observable ocean state into generic, task-blind embeddings, do climate-critical integrals like the AMOC come along for free?* The design is deliberately strict: the embedding model never sees any transport data; every read-out is a linear (ridge) regression, so any skill must already be present, linearly accessible, in the representation; and every claim is cross-validated in blocks that respect the autocorrelation of the climate system. The project also serves as a live test of a foundation-model hypothesis (§6): that a single representation trained on *all* the data — every ocean, every channel — serves specific regional tasks at least as well as one trained only on the task’s own region.

## 2 Terminology

Terms are used throughout in the following precise senses.

**Channel** one physical input field: e.g. surface current speed, mixed-layer depth, temperature at 700 dbar, zonal wind stress. The data tensor currently has 14 channels (Table 1).

**Pixel** one  $1^\circ \times 1^\circ$  ocean grid cell. A *pixel-month* is one pixel at one month — the atomic sample.

**Channel space** the space of the physical fields themselves. An error “in channel space” is an error in predicted physical values (normalized units), directly comparable with a persistence forecast of the same fields.

**Embedding / embedding space** the 64-dimensional vector  $z$  the encoder produces for a pixel-month, and the space of such vectors. An error “in embedding space” compares predicted to actual  $z$ ; it is an internal diagnostic, since  $z$  has no physical units.

**Codec** the stage-1 encoder–decoder (§4); it encodes a pixel-month’s channels to  $z$  and can decode  $z$  back to channels. “Codec” emphasizes that it is a learned compression.

**Anomaly space** all dynamic channels are expressed as departures from that pixel’s own monthly climatology (computed over training years only), then normalized. This removes everything predictable from location and calendar alone — see *climatology* below.

**Climatology** the long-term mean for a given pixel and calendar month. As a forecast, “climatology” predicts zero anomaly — the no-skill floor every real forecast must beat.

**Persistence** the forecast that the anomaly stays exactly as it is now. Hard to beat at short range; decays into a poor forecast at long range. *Damped persistence* multiplies the anomaly by its own lag- $h$  autocorrelation — the fair statistical baseline.

**Stage 2 / temporal model** a causal transformer over a pixel’s sequence of frozen embeddings that predicts next month’s embedding. “Frozen” means the codec’s weights are not updated by stage-2 training, so improvements are attributable.

**Probe** a ridge (linear,  $L_2$ -regularized) regression from embeddings to a target series. A probe measures what a representation *contains*; it is deliberately too weak to add modelling power of its own.

**Section** the row of grid cells nearest a mooring array’s latitude, clipped to the array’s longitude span (e.g. RAPID: 26.5°N, 80°W–13°W). Probes read the mean embedding along the target’s section.

**Holdout years** entire calendar years (2009, 2017, 2023) excluded from all training; with a held-out mid-Atlantic longitude band they form the blocked validation splits. Random splits are never used — adjacent months are too correlated to count as independent.

**k-fold (year-blocked)** every calendar year takes one turn as the test set while the probe is fit on the others; the assembled out-of-fold predictions are scored once. Confidence intervals come from a bootstrap that resamples whole years (a *block bootstrap*).

**chan%** held-out next-month prediction error reduction versus persistence, in channel space, in anomaly space:  $100(1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{pers}})$ . Comparable only within one channel set, since the persistence baseline moves when channels change.

**MSSS** mean squared skill score,  $1 - \text{MSE}/\text{MSE}_{\text{ref}}$  [6]; our rollout reports it against climatology, persistence, and damped persistence.

**ACC** anomaly correlation coefficient: the Pearson correlation between predicted and observed anomalies at a given forecast horizon. The forecasting literature’s lingua franca;  $\text{ACC} = 0.5$  is the conventional “useful skill” line.

**RAPID / FC / MOVE / OSNAP / SAMBA** the five transport truth series: the RAPID array at 26.5°N (2004–, monthly), the Florida Current cable at 27°N (1982–, daily), MOVE at 16°N (2000–2022), OSNAP in the subpolar gyre (2014–2022, monthly), SAMBA at 34.5°S (2009–2017, anomalies). None is ever an input.

### 3 Data

The tensor (Table 1) is built entirely from credential-free public archives. Two properties matter most. First, *missingness is information*: a pixel-month’s unobserved channels enter the encoder as learned “missing” tokens rather than imputed values, so the representation knows what the observing system knew. Second, everything is in anomaly space (§2) — the founding negative result of the project was that a model trained on raw fields wins reconstruction by memorizing the seasonal cycle while learning nothing about change.

The five truth series (RAPID, FC, MOVE, OSNAP, SAMBA) ride along as never-input targets: 240 RAPID months plus 792 additional truth months across four other latitudes.

| channel(s)                   | source               | record | role                 |
|------------------------------|----------------------|--------|----------------------|
| surface current speed, MLD   | GLORYS reanalysis    | 1993–  | dynamics             |
| T at 10/200/700/1500 dbar    | Roemmich–Gilson Argo | 2004–  | density structure    |
| S at 10/200/700/1500 dbar    | Roemmich–Gilson Argo | 2004–  | density structure    |
| wind stress $\tau_x, \tau_y$ | NCEP R1              | 1948–  | Ekman forcing        |
| SST, precip climatologies    | OISST, GPCP          | fixed  | (measured inert; §8) |
| monthly SST (staged, ch. 15) | OISST v2.1           | 1981–  | thermal memory       |

Table 1: Channels of the data tensor: monthly,  $1^\circ$ , 1993–present. The pilot window was the North Atlantic ( $100^\circ\text{W}$ – $20^\circ\text{E}$ ,  $0$ – $70^\circ\text{N}$ ; 5,787 ocean pixels); as of 2026-08-07 the default window is global (44,964 pixels, 186.7M observed values). Missing data (Argo before 2004, polar gaps) is represented explicitly, not imputed.

## 4 Methods

**Stage 1 — the codec.** A per-pixel masked autoencoder ( $\sim 0.92\text{M}$  parameters): given one pixel-month’s channels (some masked, some genuinely missing), plus latitude/longitude and month-of-year encodings, it produces  $z \in \mathbb{R}^{64}$  from which a decoder must reconstruct the masked channels *and* predict spatial and temporal neighbours. Trained on all pixel-months outside the holdout blocks; 30k steps.

**Stage 2 — the temporal model.** A causal transformer (current runner default  $d_{\text{model}}=96 \times 3$  layers, 0.35M parameters; capacity sweep in Fig. 4) over each pixel’s 24-month embedding sequence, predicting  $z_{t+1}$ ; scored after decoding to channel space. The codec stays frozen.

**Probes.** Ridge regressions from section-mean embeddings to each truth series, deseasonalized, with the regularization strength chosen on a training tail — never on test data. Verification is year-blocked k-fold with block-bootstrap confidence intervals.

**Rollout.** For horizons  $h = 1 \dots 12$ : initialize with true embeddings, predict one month, feed the prediction back, repeat — scored in channel space against climatology, persistence, and damped persistence, and probed for the AMOC at each horizon.

## 5 Metrics and their statistical power

Full definitions are in the glossary; the reliability ordering is the substance. **chan%** is decision-grade: seed spread  $\pm 0.5$  points, so differences  $\geq 2$  points are real. The **fixed-holdout probe**  $r$  (36 held-out months, effective sample  $\approx 15$  after autocorrelation) has a 95% CI of  $\pm 0.57$  — a compass, never a verdict; the same configuration has drawn 0.09 and 0.33 on two seeds. The **year-blocked k-fold** tests all 240 RAPID months once each ( $\text{SE} \approx 0.10$ ) and is the only probe number we argue from. The most instructive failure: a  $d_z=128$  codec set the all-time fixed-holdout record (0.528) while the k-fold read 0.166 — the coarse instrument flatters widest exactly where overfitting is easiest.

## 6 Related work

Three literatures border this project (full survey: repository file `m1/LITERATURE.md`). *AMOC reconstruction from observables*: regression methods reach  $r \approx 0.95$  against RAPID only after 18-month low-pass filtering and in-sample calibration [1, 2]; the machine-learning line trains on simulations and reports up to  $r \approx 0.98$  inside a laminar state estimate [3] but drops to explained variance of 0.44–0.74 in eddy-resolving models [4, 5] and is essentially never applied to real RAPID data. Our  $r = 0.60$  — monthly, deseasonalized, real data, blocked cross-validation, no RAPID-derived inputs — appears to be the first of its exact kind, and sits where the eddying simulations predict it should. *Forecast verification*: we adopt the decadal-prediction conventions [6] (MSSS, ACC, damped persistence). *Foundation models*: masked pretraining in anomaly space with frozen-backbone evaluation echoes Prithvi WxC and the frozen-decoder evaluation philosophy of [7]; ridge probes from frozen embeddings to integrated

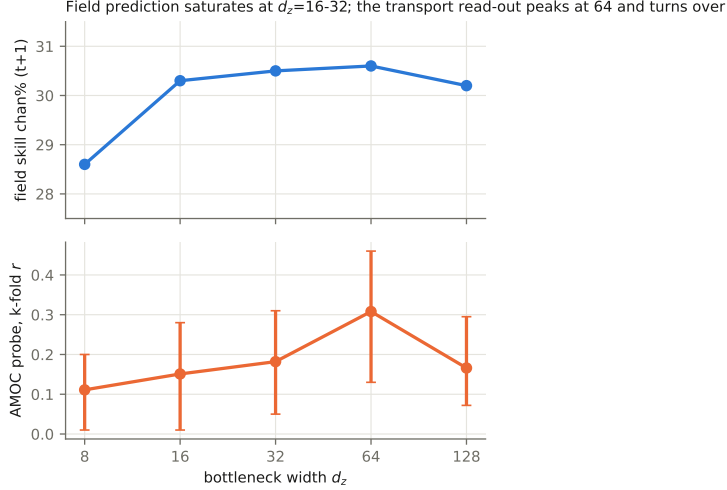


Figure 1: Bottleneck width  $d_z$  versus its two customers, on the 12-channel North-Atlantic tensor at matched training steps. Field prediction (top) saturates by  $d_z=16-32$ ; the AMOC probe (bottom, with 95% block-bootstrap CIs) rises to a peak at  $d_z=64$  and turns over at 128 — 128 ridge features against  $\sim 220$  truth months per fold overfits.  $d_z=64$  is the working setting.

circulation indices, and missingness-as-information over the historical observing system, appear to be unoccupied territory.

## 7 Results

### 7.1 The bottleneck: what width buys

Fig. 1: the transport read-out wants more width than field reconstruction rewards, but not unboundedly — the turnover at  $d_z=128$  is a truth-data limit, not a representation limit.

### 7.2 The probe ranking, and what the wind already knows

Fig. 2 is the project’s headline and its conscience in one picture. The wind channels moved the probe more than every architecture decision combined ( $0.31 \rightarrow 0.60$ ); the global codec matches the regional one exactly ( $0.602$  vs  $0.604$  — the generic-embedding hypothesis holds at fixed capacity); and the wind-only baseline at  $0.53$  says most monthly skill is Ekman physics. The embedding’s monthly increment over pure wind is  $+0.07$ ; its distinctive contributions are the low-frequency component (18-month low-passed  $r = 0.64$ ) and event anatomy (Fig. 5). In Sverdrups: RMSE  $2.23$  Sv against a target standard deviation of  $2.79$  Sv. On the four new truth series the monthly probes are null (FC  $+0.10$ , MOVE  $+0.07$ , OSNAP  $+0.13$ , SAMBA  $-0.07$ ; all CIs include zero), but FC and MOVE show low-passed correlations of  $0.35/0.32$  with *negative* wind-only baselines — slow, non-Ekman signal the embedding carries.

### 7.3 How far ahead the ocean can be predicted

Fig. 3: feeding predictions back month after month, the anomaly field stays predictable beyond a year (MSSS  $+0.11$  at  $h=12$ ; ACC  $0.36$ ). The decay is decorrelation, not amplitude damping (the forecast-to-observed amplitude ratio holds near  $0.5$  throughout). The AMOC read-out on rolled embeddings, by contrast, dies after about a season — transport forecasting at range remains open, consistent with its wind-driven monthly variance.

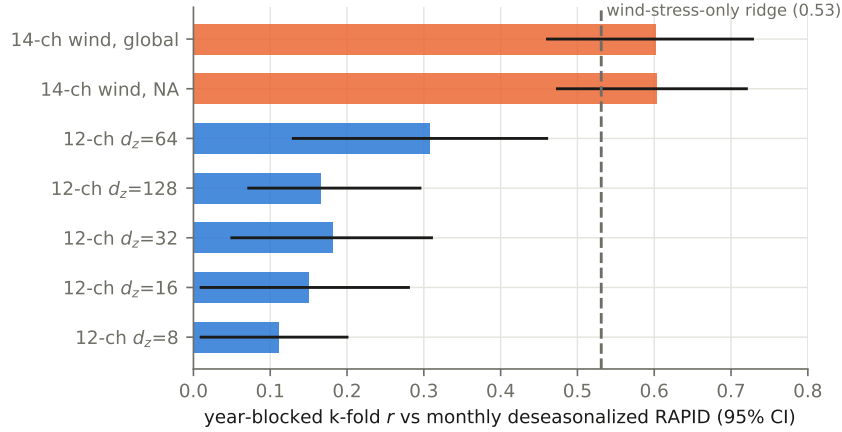


Figure 2: Year-blocked k-fold correlation with monthly deseasonalized RAPID transport, all codec generations, 95% CIs. Adding wind stress (orange) nearly doubled the defensible probe; training globally (bottom bar) cost the Atlantic read-out nothing. Dashed line: a ridge on section-mean wind stress alone, no embedding — the Ekman baseline.

## 7.4 Stage-2 capacity and the training curves

## 7.5 Case study: the winter 2009–10 collapse

# 8 The static-channel lesson

An ablation found the two climatology input channels contribute *exactly nothing* to any metric: a static map is a function of position, the encoder receives position, and so the channel carries no information the model does not already have. The general rule — a channel is worth what position and season cannot explain of it — now gates every data-acquisition decision. Climatological knowledge is still essential, but it enters as preprocessing arithmetic (the anomaly transform), never as a feature the network must carry through its bottleneck.

# 9 Limitations and open questions

(1) The wind-only baseline bounds honest claims: monthly AMOC skill is mostly Ekman; the embedding’s added value concentrates at low frequency and in event structure. (2) Absolute k-fold values are mildly optimistic — the codec saw non-holdout months’ *fields* during self-supervised pretraining (transports were never seen by anything); codec-to-codec comparisons are unaffected. (3) The rollout’s horizon is measured only to the holdout-year boundary; the crossover to no skill lies beyond 12 months and is not yet located. (4) Transport forecasting at range is unsolved. Next levers, in expected-value order: extending the time axis to 1982 (monthly SST and wind exist pre-GLORYS; the Florida cable’s 1982–92 decade becomes truth), the staged monthly-SST channel, codec growth now that the global tensor moves its data anchor to  $\sim 9M$  parameters, and label-efficiency curves over the 792 new truth months.

# References

- [1] E. Frajka-Williams (2015). Estimating the Atlantic overturning at  $26^\circ N$  using satellite altimetry and cable measurements. *GRL* 42.
- [2] A. Sanchez-Franks et al. (2021). A dynamically based method for estimating the AMOC at  $26^\circ N$  from satellite altimetry. *Ocean Science* 17.
- [3] A. Solodoch et al. (2023). Machine learning-derived inference of the meridional overturning circulation from satellite-observable variables. *JAMES* 15.

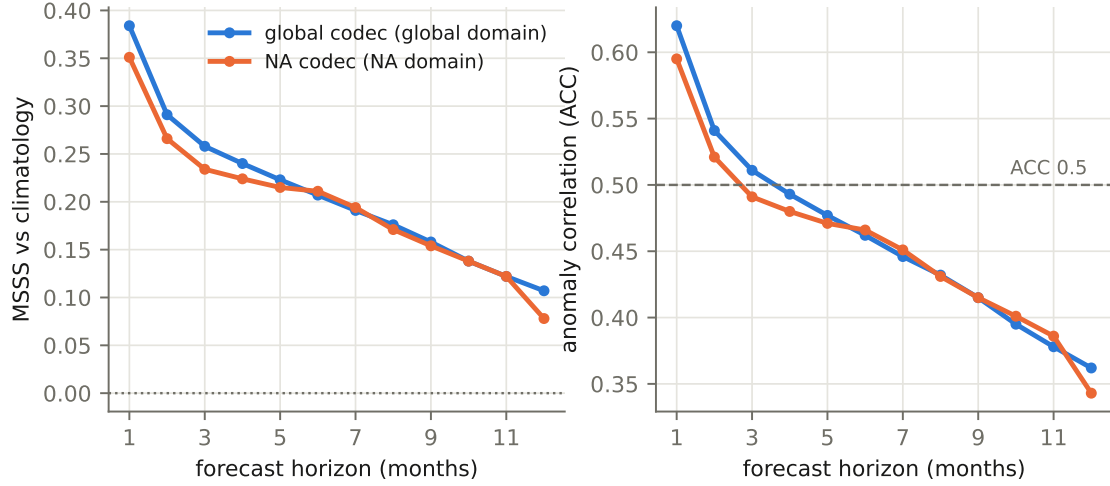


Figure 3: Autoregressive rollout skill versus forecast horizon for the global codec (blue, global domain) and the regional codec (orange, North-Atlantic domain; domains differ, so compare shapes more than levels). Left: MSSS against climatology — still positive at 12 months. Right: anomaly correlation, crossing the conventional 0.5 useful-skill line near four months. Damped persistence (not shown) is beaten at every horizon by both.

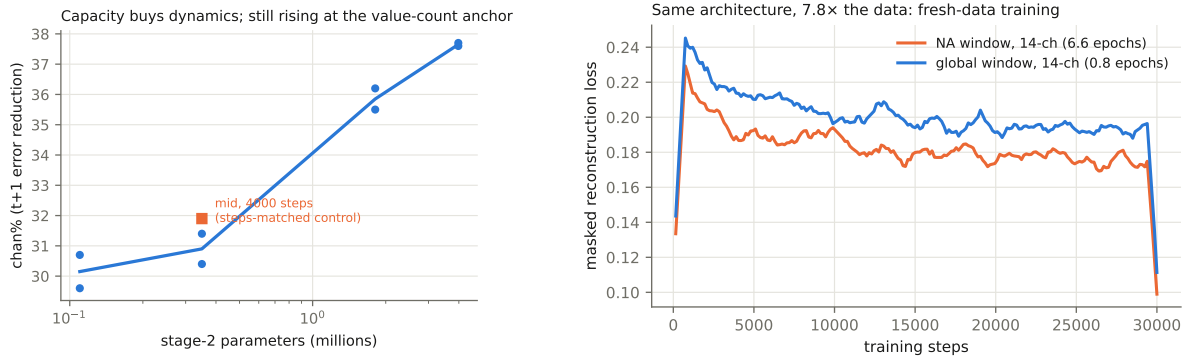


Figure 4: Left: stage-2 capacity sweep (both seeds shown per size) — width $\times$ depth buys real dynamics, still rising at 4M parameters; the steps-matched control (orange square) separates capacity from training length. Right: codec training curves — the global run sees 7.8 $\times$  the data in the same 30k steps (0.8 epochs, fully fresh data).

- [4] Y. Meng et al. (2024). Circumpolar transport and overturning strength inferred from satellite observables using deep learning. *JAMES* 16.
- [5] E. Wölker et al. (2025). Estimating the AMOC from Argo profiles with machine learning trained on ocean simulations. *Ocean Science* 21.
- [6] L. Goddard et al. (2013). A verification framework for interannual-to-decadal predictions experiments. *Climate Dynamics* 40.
- [7] F. Lehmann et al. (2026). Frozen foundation-model backbones with shallow decoders as an evaluation standard. *JGR-MLC*.

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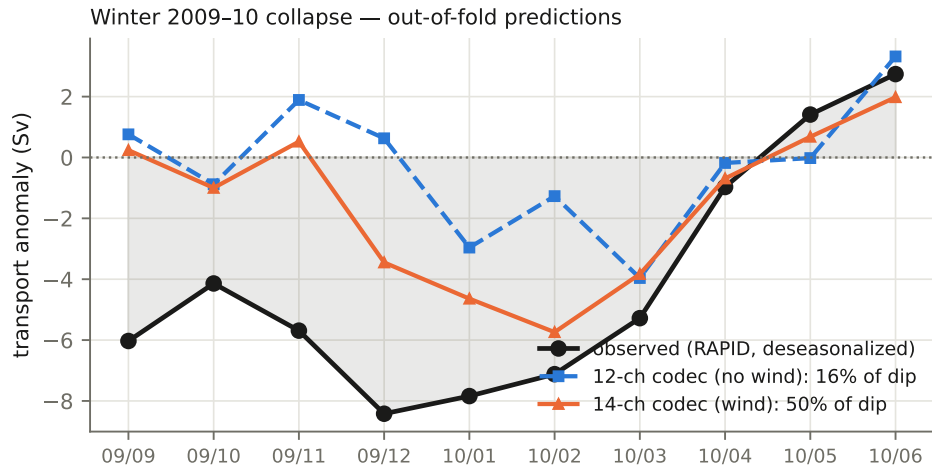


Figure 5: The sharpest event in the RAPID record, predicted out-of-fold. Without wind channels (blue) the model captures 16% of the dip amplitude; with them (orange), 50%, with the deepest month read at  $-5.7\text{ Sv}$  against  $-7.1$  observed. The November onset — triggered inside a month — stays invisible to monthly data.